

Konrad Odell

Healthcare Data Platforms & Agentic AI Systems

I've been shipping software for twenty years and setting technical direction since 2018. I joined Moxe Health, a HIPAA-regulated clinical data exchange, in 2022 and have been on its architecture team since 2024. The work is data platform and integration: Databricks lakehouse and Unity Catalog governance, FHIR pipelines, and for the last two years, evaluating LLM and agent approaches against real clinical and billing workloads: proofs of concept that had to prove their value before anyone built on them, and a clinical document generation pipeline.

On my own time I built and run ElectricityFinder, which helps people in Texas find an electricity plan. It runs an LLM extraction pipeline in production, with checks at ingest that reject bad extractions before they can be priced, plus the compliance work a regulated market requires. I also maintain the tooling I build with: MCP servers, a library of agent skills, and evaluation harnesses that decide what is allowed to ship.

The way I build with coding agents is a worker and verifier loop: one agent does the work, a separate one with fresh context has to prove it runs and does the thing before I accept it, and the ones that can't be verified don't ship. Most of my tooling exists to make that loop cheap.

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Work **Moxe Health**

● Staff Engineer, Architecture

Jan 2024 – Present

Architect for a B2B healthcare data interchange platform (HIPAA-regulated clinical data at scale), focused on data-platform capability, applied-AI evaluation, and cross-team technical coordination.

- Identified, scoped, and validated applied-AI opportunities against real clinical and financial workloads, including a multimodal LLM proof of concept that recovered \$300K in invoices that would otherwise have taken manual review to find
- Published a Claude Code skill that diagnoses production incidents from Grafana logs and traces, including a Jira template that requests the production observability access it needs; adopted by multiple teammates
- At an executive's request, evaluated moving a production document-ingestion LLM pipeline to another model provider: benchmarked alternatives on sampled documents and documented the migration path
- Designed a FHIR-to-template clinical document generation pipeline
- Built a proof of concept that turns FHIR chart data into analyst-ready extracts, validated end-to-end against a simulated FHIR harness
- Partnered on Databricks implementation including Unity Catalog governance, lakehouse architecture, and self-service tooling for data analysis
- Coordinated architecture reviews for a large-scale clinical lakehouse migration off a legacy external provider
- Documented module boundaries and state diagrams for monolith decomposition planning on a core clinical data processing system
- Coordinated architecture decisions across data platform, clinical retrieval, and revenue-operations functions, presenting tradeoffs to both engineers and senior leadership
- Driving canary deploys, smoke tests, and auto-rollback as first-class patterns in the new platform architecture (in progress)
- Coached senior engineers on architectural thinking and mentored across disciplines

● Senior Software Engineer / Team Lead

Sep 2022 – Jan 2024

Led the consent-management service team on a HIPAA-regulated B2B clinical data interchange platform; owned delivery and deploy safety, and co-owned the service architecture.

- Led consent-management service team (Java, Spring Boot, NATS) with zero-downtime blue/green deployment
- Ran the team's delivery cadence and code reviews

Senior Software Engineer II

Cognizant Softvision → Discover Financial Services

Aug 2021 – Sep 2022

Built card-tracking API ingesting USPS and internal events on Spring Boot microservices in a regulated financial-services environment.

- Introduced contract-test pipeline (WireMock + Pact) to catch breaking API changes pre-production
- Partnered with DevOps on Helm-based release process, reducing deploy time from 40 to 8 minutes

- Mentored junior engineers on testing practices and senior engineers on cross-team communication

Senior Software Engineer → VP of Research

Nearstar Inc.

Sep 2010 – Aug 2021

Legacy modernization, Docker adoption, CI/CD implementation, and performance optimization; grew from senior IC to leading technical direction as VP of Research (2018-2021).

- Dockerized monolith suite enabling per-customer configuration and a SaaS pivot
- Optimized multithreaded PDF loader during COVID spike: +260% throughput, backlog cleared in 36 hours
- Designed zero-downtime patch mechanism using Jenkins-orchestrated blue/green deployment
- Migrated Windows servers to Linux (MariaDB), enabling zero-downtime deployments and improved performance

Software Developer

ClubCorp

Jun 2001 – Jul 2008

Mainframe to Oracle migration; built authentication framework with salted password hashing.

Education

UT Dallas

B.S. in Computer Science

Projects

ElectricityFinder

- LLM plan-document extraction pipeline with checks at ingest: extractions that fail cross-contamination or confidence checks are rejected or flagged before they can be priced
- Full stack solo: Cloudflare Workers + D1, React/TypeScript frontend, Smart Meter Texas usage integration
- Registered PUCT electricity broker (BR260062); PUCT disclosure, CAN-SPAM, and TDPSA retention compliance implemented and tested
- ~5,000 automated tests, deploy-time bundle invariants, and post-deploy content smoke checks that assert rendered results, not just HTTP status

LLM Extraction

Cloudflare Workers

D1

TypeScript

Regulated Domain

Product Ownership

Agentic Engineering Estate

- Reusable building blocks: a library of 68 agent skills (usage-instrumented and pruned by actual fire-count), 45 deterministic hooks, MCP servers and tool adapters in daily production use, and sub-agent/workflow orchestration patterns
- Custom eval harnesses rather than vibes: an own-commit SWE-bench-style benchmark for scoring local and frontier models on real repository work, plus adversarial-eval and domain-specific benches. Caught a model reading the answer from git history, closed the leak, and re-audited all 122 prior runs
- Deterministic orchestration with worker/verifier separation: unattended overnight agent runs fixed 8 ElectricityFinder bugs, each re-verified from a clean checkout by a separate agent; deploys stayed human-approved
- Belief-state engine pairing LLM extraction with deterministic code judges, over a hybrid lexical + semantic search layer
- Multi-provider routing in practice (Anthropic, AWS Bedrock, Google, and local open-weight models behind one gateway), plus local serving and LoRA fine-tuning on consumer hardware

Claude Code

MCP

Multi-Agent Orchestration

Eval Harnesses

LLM-as-Judge

Python

llama.cpp

SchoolBrain

PostgreSQL

OAuth

Gemini

FastAPI

Next.js

RecipeBrain

FastAPI

PostgreSQL + pgvector

Embeddings

Publications

I Built a Factory Control Plane for AI Agents in One Afternoon

Published by konradodell.com

Mar 2026

A human control surface over a multi-phase agent pipeline (planner, worker, reviewer, verifier, monitor, reporter), built in one Claude Code session; argues the deterministic layer is the control plane, not scaffolding to discard.

Stop Getting Attached to Your AI Tooling

Published by konradodell.com

Feb 2026

Coordinating six parallel Claude Code sessions with an orchestrator/worker file protocol, and keeping that tooling disposable as the models change.

The Rutter Pattern: AI Memory Through Continuous Refinement

Two-tier agent memory: per-session logs distilled into durable patterns only once they prove themselves.

Skills

Applied AI & Agents

Production LLM pipelines

Multi-agent orchestration

AI coding agents as production build tools

MCP servers & tool integrations

Reusable agent skills / sub-agents

Retrieval & hybrid semantic search

Multi-provider model routing
(Anthropic, AWS Bedrock, Google,
local open-weight)

Local inference & LoRA fine-tuning

Evaluation & Reliability

Custom eval harnesses

LLM-as-judge gates

Adversarial evaluation

Worker/verifier separation

Deploy-time content smoke tests

Mutation & contract testing

Healthcare & Regulated Domains

HIPAA

FHIR

quality measures

Clinical data processing

PII governance

PUCT / CAN-SPAM / TDPSA

Languages

Python

TypeScript

Java

SQL

JavaScript

Bash

Data Platforms

Databricks (Unity Catalog, Genie,
Lakehouse)

Iceberg

DuckDB

Vector databases

Cloud & Distributed Systems

AWS (API Gateway, Lambda, EKS,
RDS, S3)

Cloudflare (Workers, D1, Tunnels)

Event-driven architecture (NATS)

Enterprise integration

Data pipelines

DevOps

Docker

Kubernetes

GitHub Actions

Working familiarity

Trino

dbt

HL7 v2

Supabase

